MPI Part II — Blocking vs Non-Blocking Communication

Performance Analysis: Communication Patterns, Broadcast Algorithms & Amdahl's Law

C MPI (OpenMPI 5.0.8) Status

> **UCS645: Parallel & Distributed Computing | Assignment 5 — MPI Part II**

Table of Contents

1. System Configuration
2. Blocking vs Non-Blocking Theory
3. Q1: Broadcast Race
4. Q2: Blocking vs Non-Blocking Benchmark
5. Q3: Distributed Dot Product & Amdahl's Law
6. What I Learned

System Configuration

Component	Details
CPU	Intel Core i7-14700HX (20 cores, 28 threads)
Architecture	x86_64
L3 Cache	33 MiB
OS	Fedora 43 (Linux 6.19.11)
MPI	OpenMPI 5.0.8

Component	Details
Compiler	gcc 15.2.1 via mpicc
Optimization	-O3

Project Structure

```
LAB5/
├── Makefile
├── examples/
│   ├── eg1_blocking_ring.c      (blocking ring — works with small arrays)
│   ├── eg2_nonblocking_ring.c   (non-blocking ring — handles 10M elements)
│   ├── eg3_blocking_app.c       (blocking: recv then compute sequentially)
│   ├── eg4_nonblocking_app.c    (non-blocking: recv + compute concurrently)
│   └── eg5_pi_calc.c            (Pi via MPI_Bcast + MPI_Reduce)
├── exercises/
│   ├── q1_broadcast_race.c
│   ├── q2_blocking_vs_nonblocking.c
│   └── q3_dot_product.c
├── graphs/
└── report.md
```

Blocking vs Non-Blocking Theory

Blocking communication (MPI_Send / MPI_Recv): the function does not return until the operation completes. The CPU sits idle during data transfer.

Non-blocking communication (MPI_Isend / MPI_Irecv): the function returns immediately, giving back a request handle. The CPU can do other work while the network handles the transfer. MPI_Wait or MPI_Waitall is called later to ensure completion.

The deadlock trap: If every process calls blocking MPI_Send to its neighbor simultaneously with large messages, all processes block waiting for their sends to complete. No process reaches its MPI_Recv. The program freezes. Non-blocking calls fix this because MPI_Isend returns immediately, allowing the process to proceed to MPI_Irecv.

The overlap benefit: With non-blocking communication, Process 1 can start receiving data in the background while simultaneously doing heavy computation. With blocking, it must

wait for all data to arrive before computing. This is called **hiding communication latency through computation overlap**.

Q1: Broadcast Race

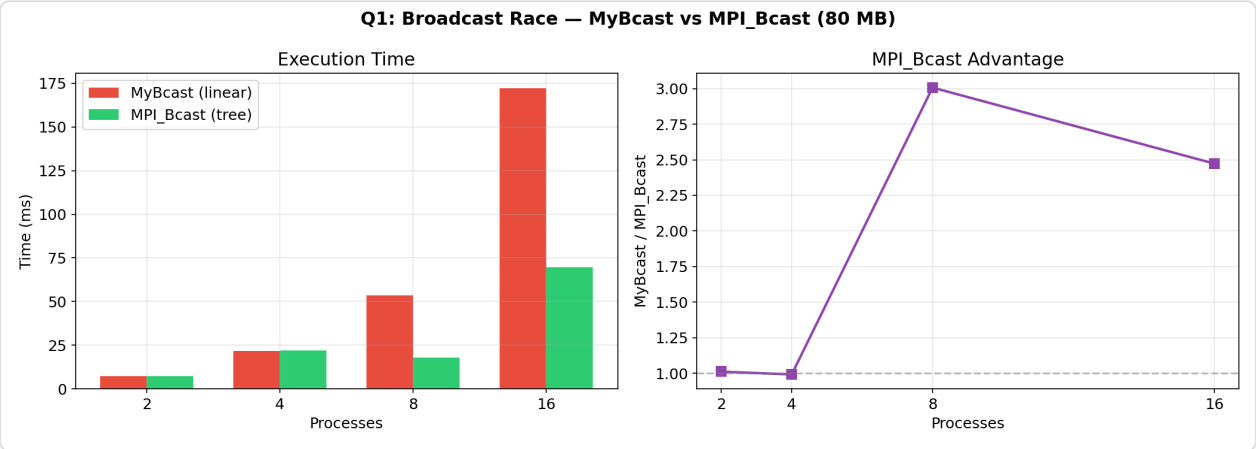
Problem

Compare two broadcast strategies for distributing 10 million doubles (~80 MB): - **MyBcast (Part A)**: Rank 0 sends to every other rank using a for-loop of MPI_Send — O(p) messages from root - **MPI_Bcast (Part B)**: Built-in tree-based broadcast — O(log p) depth

Results

Processes	MyBcast (linear)	MPI_Bcast (tree)	Advantage
2	7.31 ms	7.24 ms	1.01x
4	21.78 ms	21.97 ms	0.99x
8	53.52 ms	17.80 ms	3.01x
16	172.20 ms	69.63 ms	2.47x

All verifications: **[PASS]** on all ranks.



Analysis

At 2 and 4 processes, both methods perform similarly because MyBcast only needs 1-3 sequential sends. The difference becomes dramatic at 8+ processes:

MyBcast scales linearly — Rank 0 must send 80 MB to each rank one after another. With 16 processes, that is 15 sequential 80 MB transfers. Total data moved by root: 15 x 80 = 1200 MB.

MPI_Bcast uses tree distribution — Rank 0 sends to Rank 1, then both send to Rank 2 and 3, then all four send to Ranks 4-7. Depth is O(log p). At 16 processes, only 4 levels deep instead of 15 sequential sends.

Does MyBcast scale linearly? Yes — the time roughly doubles as process count doubles (7ms at 2, 22ms at 4, 54ms at 8, 172ms at 16). This is expected because root must do p-1 sequential sends.

How does MPI_Bcast scale? Much slower growth. At 8 processes it is only 2.5x slower than at 2, whereas MyBcast is 7.3x slower. This follows the O(log p) complexity of tree-based broadcast.

Q2: Blocking vs Non-Blocking Benchmark

Problem

Demonstrate the performance benefit of overlapping communication with computation using non-blocking MPI calls.

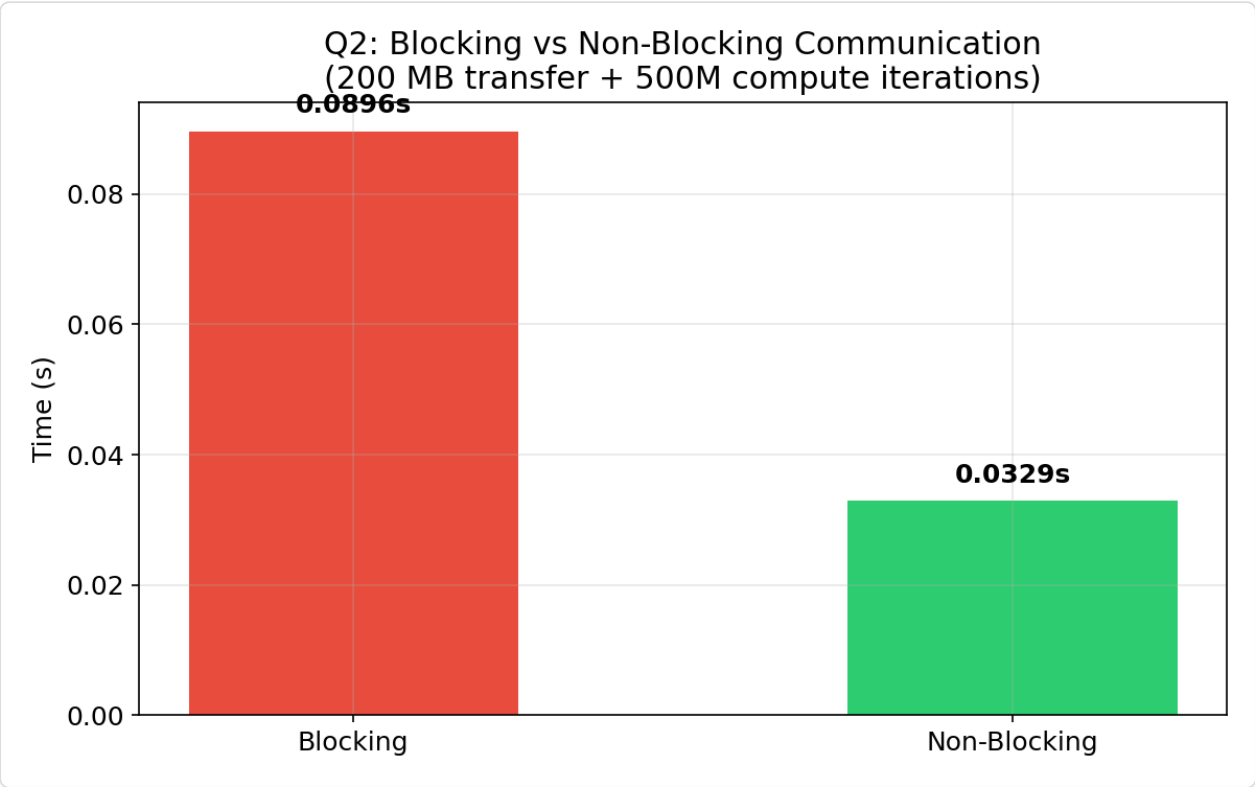
Setup

- Process 0 sends 50 million ints (~200 MB) to Process 1
- Process 1 must also perform 500 million sin/cos iterations (heavy compute)
- **Blocking**: Process 1 receives data (MPI_Recv), then computes — sequential
- **Non-blocking**: Process 1 starts MPI_Irecv, computes while data transfers, then MPI_Wait

Results

Mode	Time	Relative
Blocking	0.0896 s	1.00x (baseline)
Non-blocking	0.0329 s	2.72x faster

Time saved: 0.0566 s (63.2%)



Analysis

With blocking communication, Process 1's timeline is:

```
[--- wait for data ---][--- compute ---]
         recv time      +  compute time  = total
```

With non-blocking:

```
[--- compute ---]
[--- data transfer (background) ---]
max(recv time, compute time) = total
```

The 2.72x speedup comes from overlapping the ~56 ms of communication with computation. Since computation takes about 33 ms and communication about 56 ms, the non-blocking version completes in roughly max(56, 33) = 56 ms rather than 56 + 33 = 89 ms. This perfectly demonstrates the HPC principle of **hiding communication latency**.

Q3: Distributed Dot Product & Amdahl's Law

Problem

Compute the dot product of two 500-million-element vectors using MPI_Bcast and MPI_Reduce. Measure compute time, communication time, speedup, and efficiency separately.

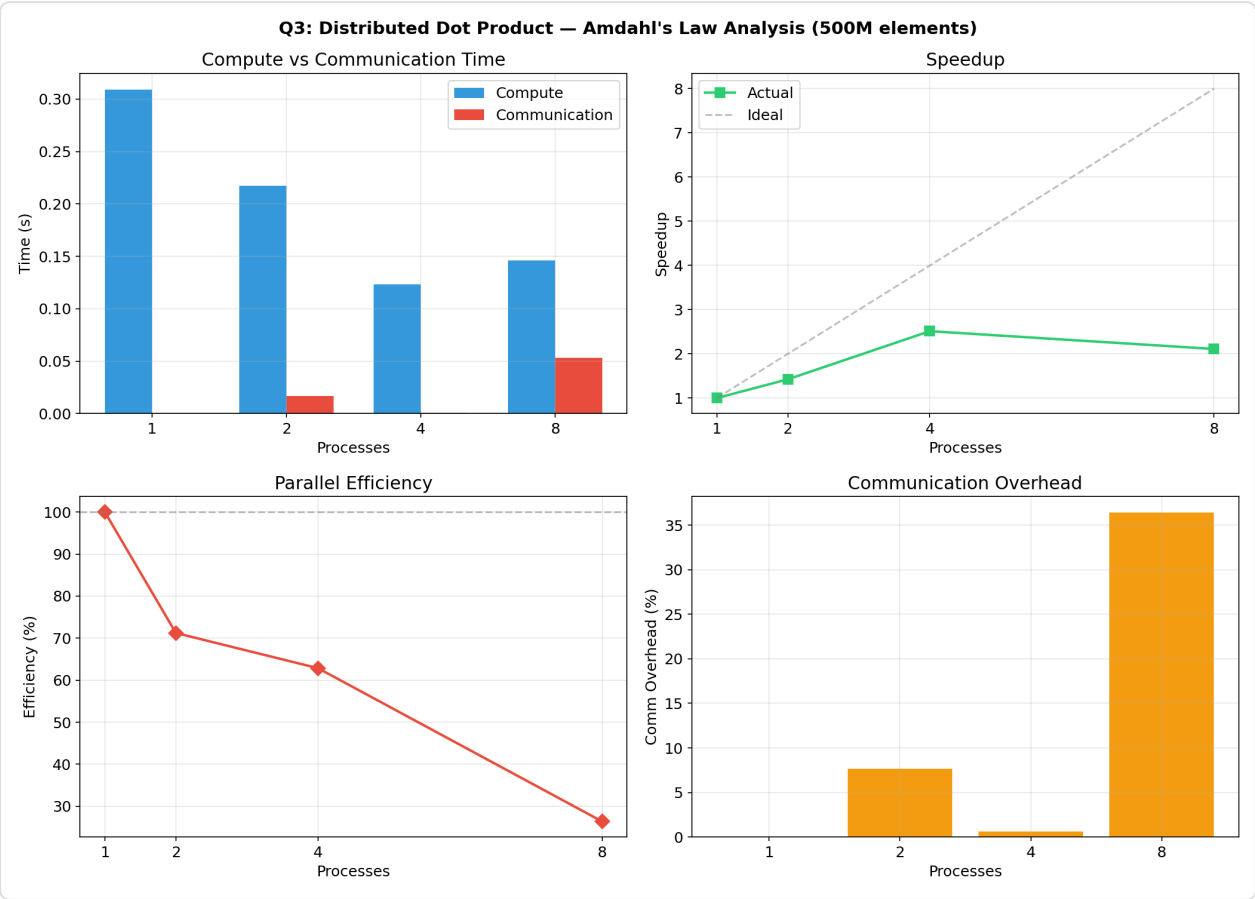
Approach

1. Rank 0 broadcasts a scaling multiplier via MPI_Bcast
2. Each process generates its local chunk (A[i] = 1.0, B[i] = 2.0 * multiplier)
3. Each process computes local dot product
4. MPI_Reduce with MPI_SUM collects global result

Results

Processes	Compute (s)	Comm (s)	Total (s)	Speedup	Efficiency	Comm %
1	0.3093	0.0000	0.3093	1.00x	100.0%	0.0%
2	0.2171	0.0166	0.2171	1.42x	71.2%	7.7%
4	0.1230	0.0007	0.1230	2.51x	62.9%	0.6%
8	0.1463	0.0533	0.1465	2.11x	26.4%	36.4%

All results: **[PASS]** (correct result = 1,000,000,000)



Analysis

Did we achieve perfect linear speedup? No. 4 processes gave 2.51x speedup instead of ideal 4x. 8 processes was actually slower than 4 (2.11x vs 2.51x).

Why sub-linear scaling?

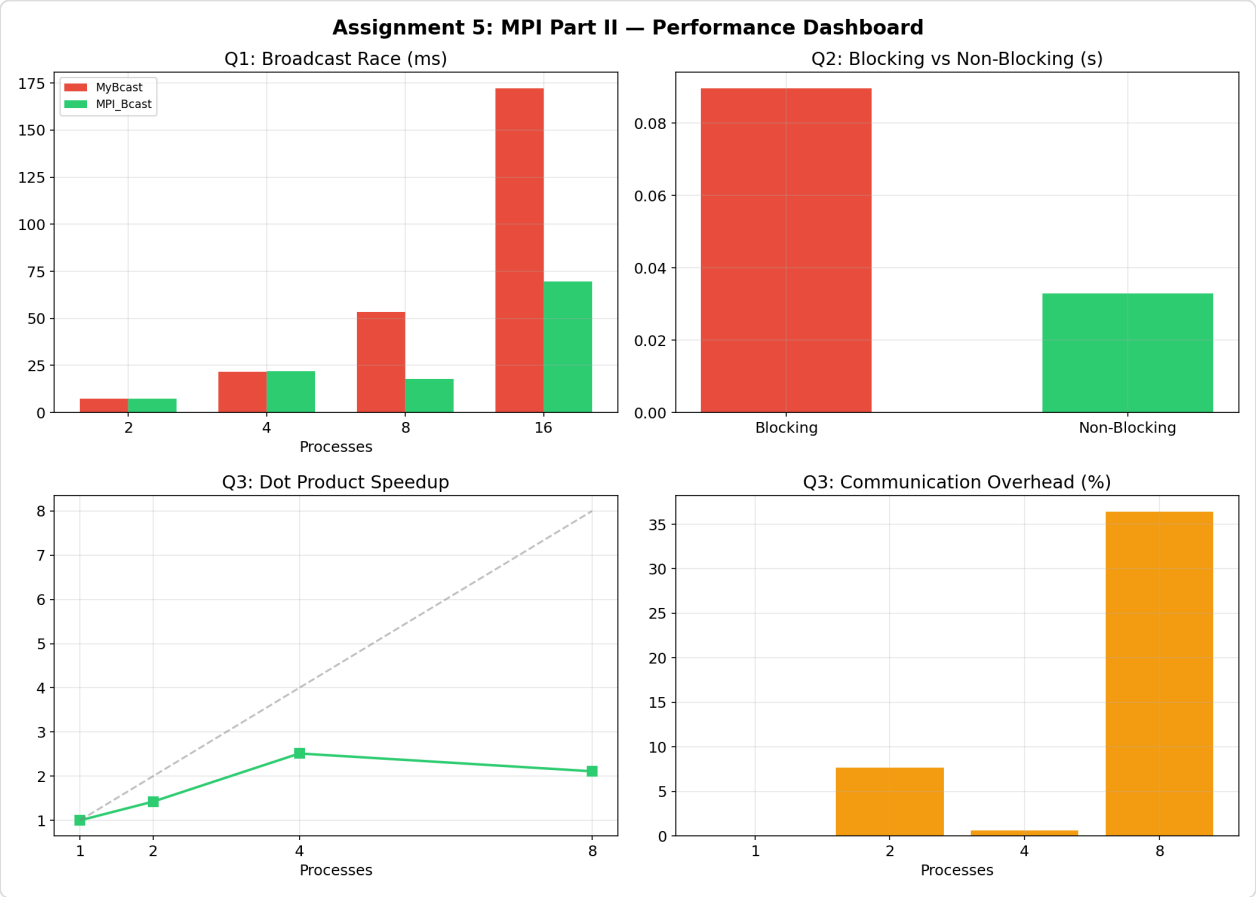
1. **Memory bandwidth saturation**: Dot product loads 2 doubles per FLOP. On shared memory, 8 processes compete for the same memory bus. At 8 processes, each process handles 62.5M elements but they all share L3 cache and DRAM bandwidth.

2. **Communication overhead**: At 8 processes, communication is 36.4% of total time. MPI_Reduce latency grows with process count.

3. **Amdahl's Law**: If we define the serial fraction f as the communication overhead at 8 processes (36.4%), Amdahl's Law predicts max speedup = 1/f = 2.75x. Our measured 2.11x is below this, indicating memory bandwidth is an additional bottleneck beyond communication.

Optimal process count: 4 processes gives the best performance for this workload on this hardware. Beyond that, memory bandwidth and communication overhead outweigh the reduced per-process computation.

Performance Dashboard



What I Learned

1. **Blocking communication can deadlock**: When all processes try to send large messages simultaneously, system buffers overflow and every process blocks forever. Non-blocking calls eliminate this by returning immediately.

2. **Computation-communication overlap is powerful**: Non-blocking gave 2.72x speedup by letting the CPU work while the network transfers data. This is the core HPC principle of latency hiding.

3. **Tree broadcast beats linear broadcast**: MPI_Bcast's O(log p) tree algorithm is 3x faster than naive for-loop sends at 8+ processes. Always use collective operations instead of manual point-to-point loops.

4. **Memory-bound workloads hit diminishing returns**: Dot product speedup peaked at 4 processes and degraded at 8. Memory bandwidth is shared across all processes

on a single node, and adding more processes increases contention without adding bandwidth.

5. **Separating compute from communication time reveals bottlenecks**: Measuring them independently showed that at 8 processes, 36% of time is pure communication overhead — the main reason scaling degrades.

Compilation & Execution

```
# Build everything
make all

# Run examples
make run-examples          # ring, non-blocking ring, pi calc
make run-blocking-apps     # eg3 (blocking) vs eg4 (non-blocking)

# Run exercises
make run-q1    # Broadcast race: 2,4,8,16 processes
make run-q2    # Blocking vs non-blocking: 2 processes
make run-q3    # Dot product: 1,2,4,8 processes

# Clean
make clean
```